

SCE-015 — STRUCTURAL COGNITIVE RUNTIME: THE TWIN OF KNOWLEDGE AND MECHANISM

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with chatGPT (OpenAI)

2026-06-19

Repository: <https://github.com/sizhet/Structural-Cognitive-Emergence-SCE>

Abstract

Structural Cognitive Emergence (SCE) has progressively revealed that cognition cannot be fully explained by accumulated knowledge alone. A child, an animal, a scientist, or a future Autonomous Structural Intelligence (ASI) system may possess different amounts of knowledge, yet all exhibit similar patterns of reasoning, question generation, gap discovery, explanation construction, and adaptive learning.

This observation suggests that cognition consists of two inseparable components:

- Structural Knowledge
- Structural Runtime

Structural Knowledge represents what is known.

Structural Runtime represents how knowledge is activated, expanded, evaluated, explained, and updated.

This work proposes a Structural Cognitive Runtime model and argues that cognition emerges not from knowledge alone, but from the continuous interaction between structural knowledge and structural runtime.

1. Introduction

Many theories of intelligence focus primarily on knowledge.

Examples include:

- facts
- concepts
- memories
- rules
- experiences

However, a fundamental question remains:

Why can two individuals possessing similar knowledge exhibit dramatically different reasoning abilities?

The answer may lie in the distinction between knowledge and mechanism.

Knowledge alone does not reason.

Knowledge alone does not ask questions.

Knowledge alone does not discover gaps.

Something must continuously operate upon knowledge.

SCE refers to this operating process as Structural Cognitive Runtime.

2. Knowledge Is Not Cognition

Consider a newborn infant.

The infant possesses very little explicit knowledge.

Yet the infant already demonstrates:

- attention
- exploration
- curiosity
- preference formation
- feedback adaptation

This suggests that cognition cannot depend entirely upon stored knowledge.

The runtime must exist before significant knowledge accumulation occurs.

In this sense:

Knowledge can be sparse.

Runtime cannot be absent.

3. Structural Knowledge

Current SCE research identifies three major structural knowledge layers.

3.1 Differential Tree

Represents:

- categories
- naming
- classification
- membership

Answers:

What is it?

3.2 Calling Graph

Represents:

- relations
- workflows
- causality
- dependencies

Answers:

What is connected?

3.3 Triggering Links

Represents:

- threshold activation
- decision triggers
- state transitions

Answers:

What happens next?

Together they form Structural Knowledge.

4. Structural Runtime

Knowledge structures alone do not explain cognition.

A runtime must continuously operate upon those structures.

Current SCE evidence suggests the following runtime sequence.

Step 1: Locate

Locate relevant nodes.

Example:

Observed Animal



Cat Node

Step 2: Expand

Expand surrounding structures.

Cat
↓
Mammal
Pet
Predator
Domestic Animal

Step 3: Trigger

Activate related behavioral and decision structures.

Hungry
↓
Food Seeking

Step 4: Score

Aggregate evidence.

Evidence
↓
Scoring Tree

Step 5: Decide

Select the strongest structural interpretation.

Step 6: Explain

Detect structural gaps.

Generate:

Why?

and

How?

Step 7: Update

Modify structures using feedback.
This completes one runtime cycle.

5. The Structural Cognitive Loop

The complete runtime can be summarized as:

Locate



Expand



Trigger



Score



Decide



Explain



Update

Or structurally:

Differential Tree



Calling Graph



Triggering Links



Scoring Tree



Decision



Gap Why



Feedback

This loop may represent the minimal cognitive cycle of Structural Intelligence.

6. Why Questions Emerge

One of the most important consequences of this runtime is the emergence of questions.

Traditional systems often treat questions as language outputs.

SCE proposes a different view.

Questions emerge when runtime encounters unresolved structures.

Examples:

Missing Node

Missing Relation

Missing Trigger

Missing Evidence

These become:

Why?

How?

What if?

What next?

Question generation is therefore a natural consequence of runtime operation.

7. Gap Why as Runtime Activity

Previous SCE studies demonstrated:

Why

=

Gap Detection

+

Gap Completion

This interpretation fits naturally into the runtime framework.

The runtime continuously compares:

Expected Structure

with

Observed Structure

Differences produce gaps.

Gaps generate explanations.

Thus Why generation is not a special language capability.

It is a runtime function.

8. Infant Learning and Runtime Primacy

Infants provide a powerful clue regarding the origin of cognition.

An infant possesses:

- limited concepts
- limited memories
- limited language

Yet exhibits:

- exploration
- adaptation
- preference formation
- attention shifts

This suggests:

Runtime precedes knowledge scale.

Knowledge accumulates later.

The runtime already exists.

Thus infant cognition may be one of the clearest windows into the

fundamental architecture of intelligence.

9. From Humans to ASI

The same runtime appears applicable across multiple forms of intelligence.

Human Learning

Knowledge

+

Runtime

Scientific Discovery

Knowledge

+

Runtime

Education

Knowledge

+

Runtime

Autonomous Coding

Knowledge

+

Runtime

Future ASI

Knowledge

+

Runtime

The difference may lie primarily in:

- scale
- speed
- depth
- feedback efficiency

rather than in fundamentally different architectures.

10. The Twin Principle

The central proposal of this work is the Twin Principle.
Cognition always consists of two inseparable components.

Structural Knowledge

and

Structural Runtime

Knowledge provides structure.

Runtime provides activity.

Knowledge provides memory.

Runtime provides cognition.

Neither alone is sufficient.

Together they produce intelligence.

Conclusion

Structural Cognitive Emergence increasingly suggests that intelligence cannot be understood solely through accumulated knowledge.

Beneath concepts, memories, categories, and relations lies a continuously operating runtime responsible for localization, expansion, triggering, evaluation, explanation, and adaptation.

This leads to a simple but powerful formulation:

Structural Cognition

=

Structural Knowledge

+

Structural Runtime

The distinction between knowledge and mechanism may represent one of the most important conceptual foundations for future studies of human

cognition, infant learning, scientific discovery, education, and Autonomous Structural Intelligence.